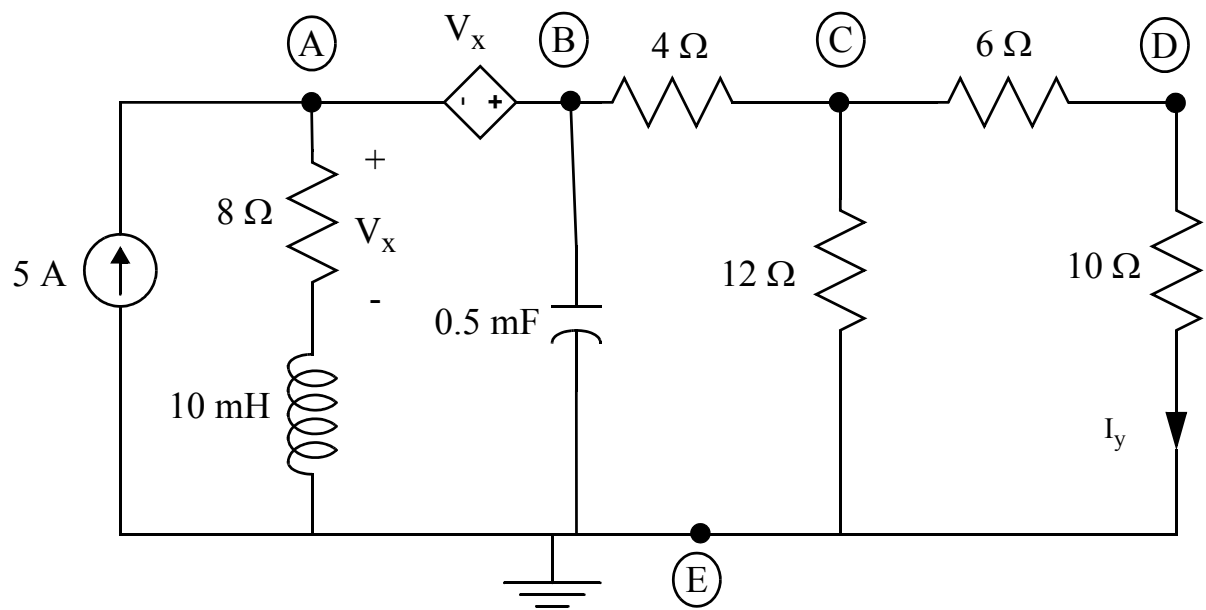


Instructions:

1. This is a CLOSED book examination.
 2. Check that your name and student id is given correctly at the top of this page. This exam has assigned seating and you should sit only where the exam paper with your name and student id is placed.
 3. Check that you have 4 questions in your paper. You should answer all 4 questions. The maximum marks for each question is given in the paper. Maximum marks for the entire exam is 40.
 4. There are 14 pages in this exam. You can write on both sides of each sheet of paper. Return all pages at the end of the exam.
 5. Each question has space below it (on the same page) and on the following two pages to write your answer.
 6. If you do not have enough space to answer a question on the three assigned pages, then continue writing where there is space and indicate where you have continued to answer this question.
 7. Clearly show all steps used in your solution. No marks will be given for numerical results unless accompanied with the correct solution method. Give units for all answers.
 8. Simple scientific calculators with up to four-line display are allowed in the exam. Calculators that are programmable, graphing or calculators with more than four-line display are not allowed.
 9. Other than the allowed calculator, you should not have any other electronic or communication devices (e.g. cell phone, laptop, etc.) with you while writing the exam. All such devices must be placed out of your reach during the exam.
-

Q1) Using *Nodal Analysis*, and without changing, simplifying, or transforming the DC circuit in any way, calculate:



- (a)

The current I_y .

(8 marks)
- (b)

The energy stored in the capacitor.

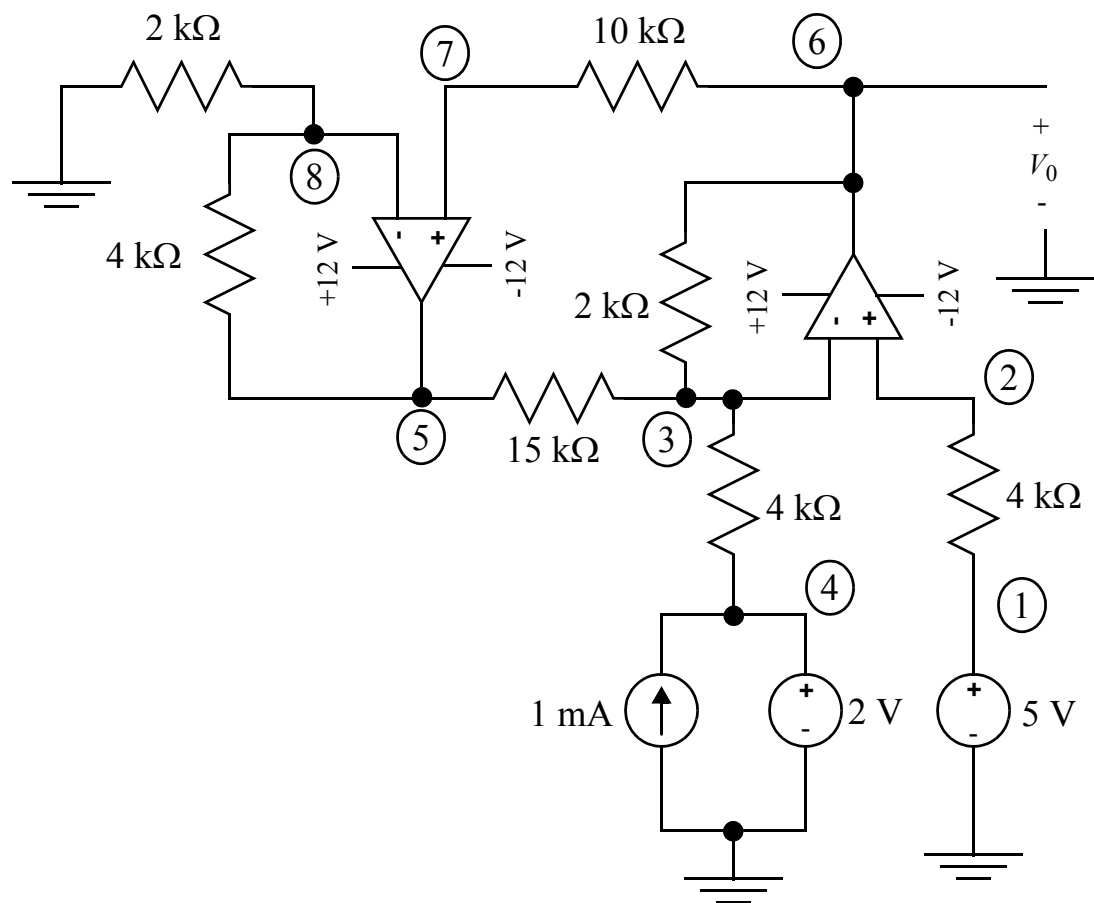
(2 marks)

Assigned page to write answer for Q1.

Assigned page to write answer for Q1.

Q2) Determine the voltage V_o in the following circuit, considering the potential saturation of the Opamps.

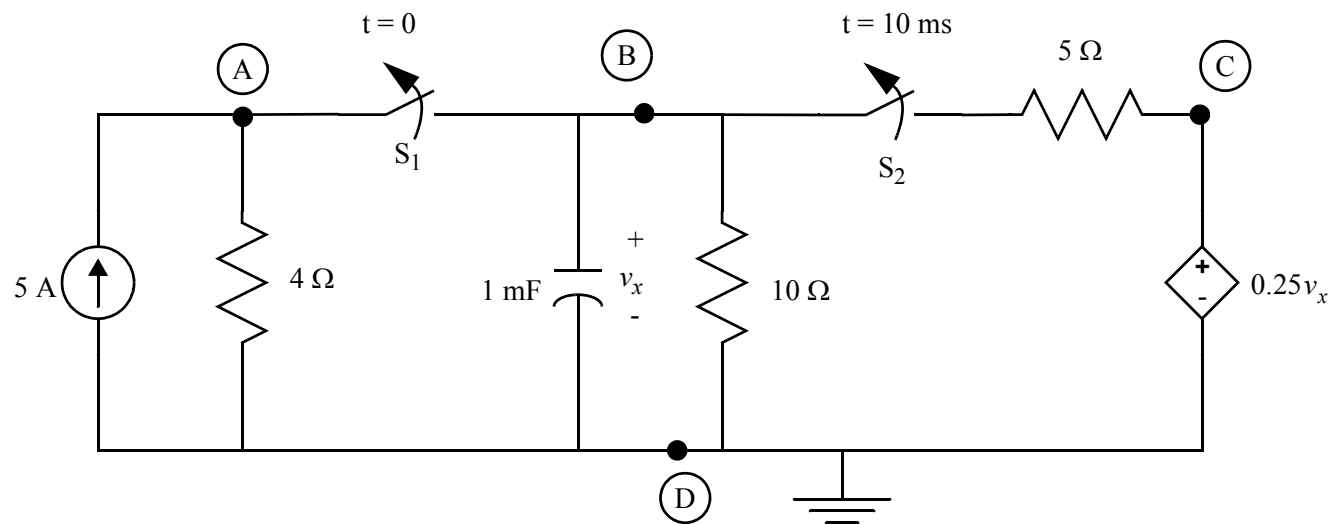
(10 marks)



Assigned page to write answer for Q2.

Assigned page to write answer for Q2.

Q3) In this question both switches are closed for a long time and circuit is thus in steady state at $t = 0$.

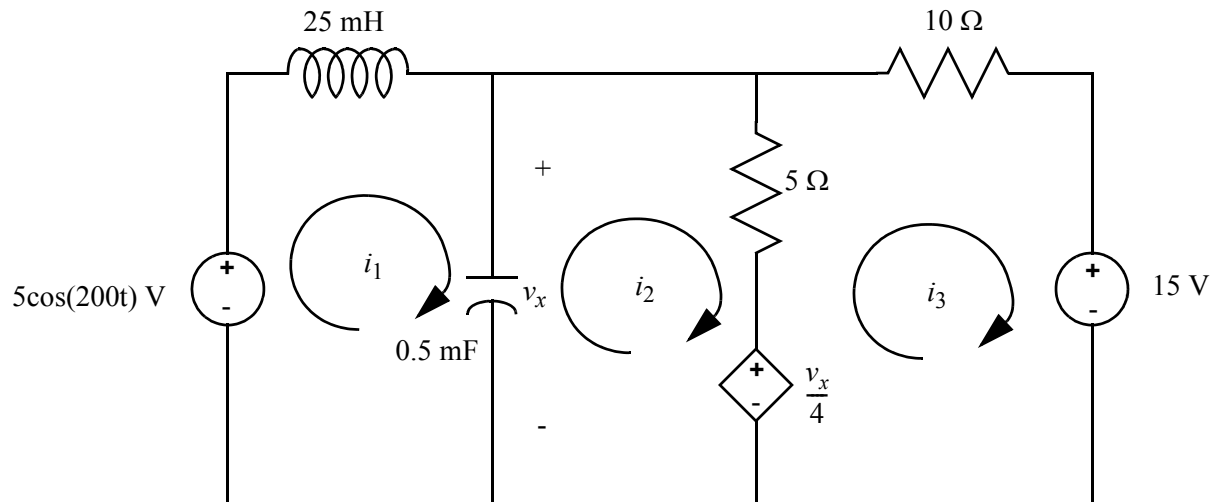


- (a) Find the initial condition on the capacitor $v_x(0)$. (2 marks)
- (b) At $t = 0$, switch S_1 opens. Calculate an expression for the capacitor voltage $v_x(t)$ for $t \geq 0$. (5 marks)
- (c) At $t = 10$ ms, switch S_2 opens. Calculate an expression for the capacitor voltage $v_x(t)$ for $t \geq 10$ ms. (3 marks)

Assigned page to write answer for Q3.

Assigned page to write answer for Q3.

Q4) Using ***Mesh Analysis***, and without changing, simplifying, or transforming the circuit in any way, calculate:



- (a) The voltage $v_x(t)$ in steady state. (7 marks)
- (b) The average real power absorbed by the 5Ω resistor. (2 marks)
- (c) The reactive power in the capacitor. (1 mark)

Assigned page to write answer for Q4.

Assigned page to write answer for Q4.

Formula Sheet

Current:	$i = \frac{dq}{dt}$	[A]	Ohm’s Law: $v = R i$ or $i = G v$ where $G = \frac{1}{R}$
Voltage:	$v = \frac{dw}{dq}$	[V]	Resistors in Series: $R_{eq} = R_1 + R_2 + R_3$
Power:	$p = \frac{dw}{dt} = v \cdot i$	[W]	Resistors in Parallel: $\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$

Voltage, Current, and Energy

Device	Voltage	Current	Stored Energy
Resistor, R	$v(t) = R i(t)$	$i(t) = \frac{v(t)}{R}$	
Inductor, L	$v(t) = L \frac{di(t)}{dt}$	$i(t) = i(t_o) + \frac{1}{L} \int_{t_o}^t v(t) dt$	$w(t) = \frac{L i(t)^2}{2}$
Capacitor, C	$v(t) = v(t_o) + \frac{1}{C} \int_{t_o}^t i(t) dt$	$i(t) = C \frac{dv(t)}{dt}$	$w(t) = \frac{C v(t)^2}{2}$

Transient Analysis

RC Circuit	$v(t) = v(\infty) + [v(t_0) - v(\infty)]e^{-\frac{t-t_0}{\tau}}; \tau = RC$
RL Circuit	$i(t) = i(\infty) + [i(t_0) - i(\infty)]e^{-\frac{t-t_0}{\tau}}; \tau = L/R$

AC Impedance

General	Resistor, R	Inductor, L	Capacitor, C
$Z = \frac{V}{I} = R + jX$ $Z \angle \theta_Z = \frac{V_m}{I_m} \angle \theta_V - \theta_I$	$R = R \angle 0^\circ$	$j\omega L = \omega L \angle 90^\circ$	$\frac{-j}{\omega C} = \frac{1}{\omega C} \angle -90^\circ$

AC Power

Complex, S [VA]	Real, P [W]	Reactive, Q [VAR]
$S = V_{rms} I_{rms}^* = P + jQ$	$P = V_{rms} I_{rms} \cos(\theta_V - \theta_I)$	$Q = V_{rms} I_{rms} \sin(\theta_V - \theta_I)$

$V_{rms} = \frac{V_m}{\sqrt{2}}; \quad I_{rms} = \frac{I_m}{\sqrt{2}}$

$R \angle \theta = R \cos \theta + j R \sin \theta; \quad a + jb = \sqrt{a^2 + b^2} \angle \tan^{-1} \left(\frac{b}{a} \right)$

$\sin(\theta) = \cos(\theta - 90^\circ); \quad -\sin(\theta) = \cos(\theta + 90^\circ); \quad -\cos(\theta) = \cos(\theta \pm 180^\circ)$